

Multiple Choice (10 marks)

1. What factors determine the characteristics of particular organisms that exist in an ecosystem.
 - (a) Temperature, humidity, and human activities
 - (b) Only soil structure and chemistry
 - (c) Only sunlight and humidity
 - (d) Sunlight, soil pH, and predator-prey relationships
 - (e) Only temperature and precipitation
2. If the activity of an enzyme is constant over a broad range of pH values, it is likely that
 - (a) the enzyme is inactive at all pH values
 - (b) no ionizing groups on the enzyme or substrate participate in the reaction
 - (c) only ionizing groups on the substrate participate in the reaction
 - (d) ionizing groups on both the enzyme and substrate participate in the reaction
 - (e) only non-ionizing groups on the substrate participate in the reaction
3. How do cyanobacteria differ from green unicellular algae?
 - (a) Cyanobacteria reduce sulfur compounds.
 - (b) Cyanobacteria have no nuclei.
 - (c) Green algae cannot reduce nitrogen compounds.
 - (d) Green algae cannot photosynthesize.
 - (e) Green algae produce cell membranes.
4. Explain why simple diffusion is not sufficient to account for the passage of all of the dissolved minerals into the cells of water plants.
 - (a) Simple diffusion can only transport minerals when they are bound to oxygen molecules
 - (b) Simple diffusion cannot occur against a concentration gradient
 - (c) Simple diffusion requires specific carrier proteins for each mineral
 - (d) Simple diffusion only works for organic compounds, not inorganic minerals
 - (e) Simple diffusion can transport minerals in any concentration
5. The pH of two lakes is measured. Lake A has a pH of 8.0; Lake B has a pH of 6.0. Which of the following statements is correct about these lakes?

- (a) Lake B is alkaline.
- (b) Lake B is 100 times more alkaline than Lake A.
- (c) The hydrogen ion concentration of Lake B is 10^{-4} M.
- (d) The pH of both lakes will decrease in response to acid rain.

True / False (10 marks)

Answer True or False

- 6. True or False: Are lower fasting plasma glucose levels at diagnosis of type 2 diabetes associated with improved outcomes?
- 7. True or False: Globulomaxillary cysts—do they really exist?
- 8. True or False: Is late-night salivary cortisol a better screening test for possible cortisol excess than standard screening tests in obese patients with Type 2 diabetes?
- 9. True or False: Chemoradiotherapy in the management of locally advanced squamous cell carcinoma esophagus: is surgical resection required?
- 10. True or False: Is vitamin D deficiency a feature of pediatric celiac disease?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the expected genotypic ratios of the offspring from the cross $GgDdww \times GgddWw$, specifically focusing on the fraction that would be homozygous recessive for all gene pairs.
- 12. Discuss how inbreeding and loss of genetic variation contribute to the vulnerability of small populations and their progression toward extinction. Provide examples to support your analysis.

End of Paper